

Image Processing Project

Feature-Based Image Processing and Reinforcement Learning for CarRacing-v3

Main contribution: image-processing-derived compact visual feature representation

Downstream evaluation: PPO and SAC on Gymnasium CarRacing-v3

Final package

Report, slides, notebooks, logs, figures, tables, and demo videos. V2 is final-facing; V1 is archived history.

Problem and project goal

Problem

Raw RGB driving observations mix visual interpretation and control. Training an end-to-end image policy is costly for a course-scale notebook workflow.

Goal

Extract compact road geometry from images, then test whether standard PPO/SAC policies can use those features to drive.

The project is evaluated with reinforcement learning, but the contribution is the visual feature pipeline.

Why image processing?

Compact state

A small feature vector is easier to train on than a $96 \times 96 \times 3$ image tensor.

Inspectable features

Road mask, ray distances, and radar-style summaries can be checked directly.

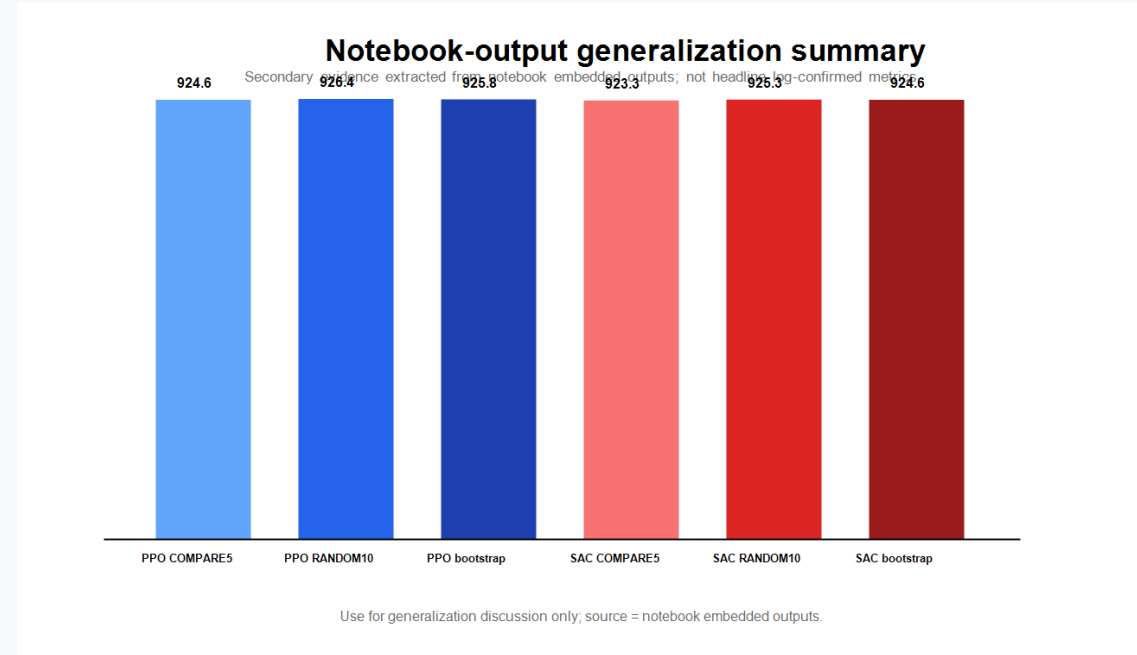
Course-scale evidence

The package can explain the visual representation without claiming a new RL algorithm.

Design tradeoff: hand-engineered features improve interpretability and speed, but can fail when the mask or local geometry is misleading.

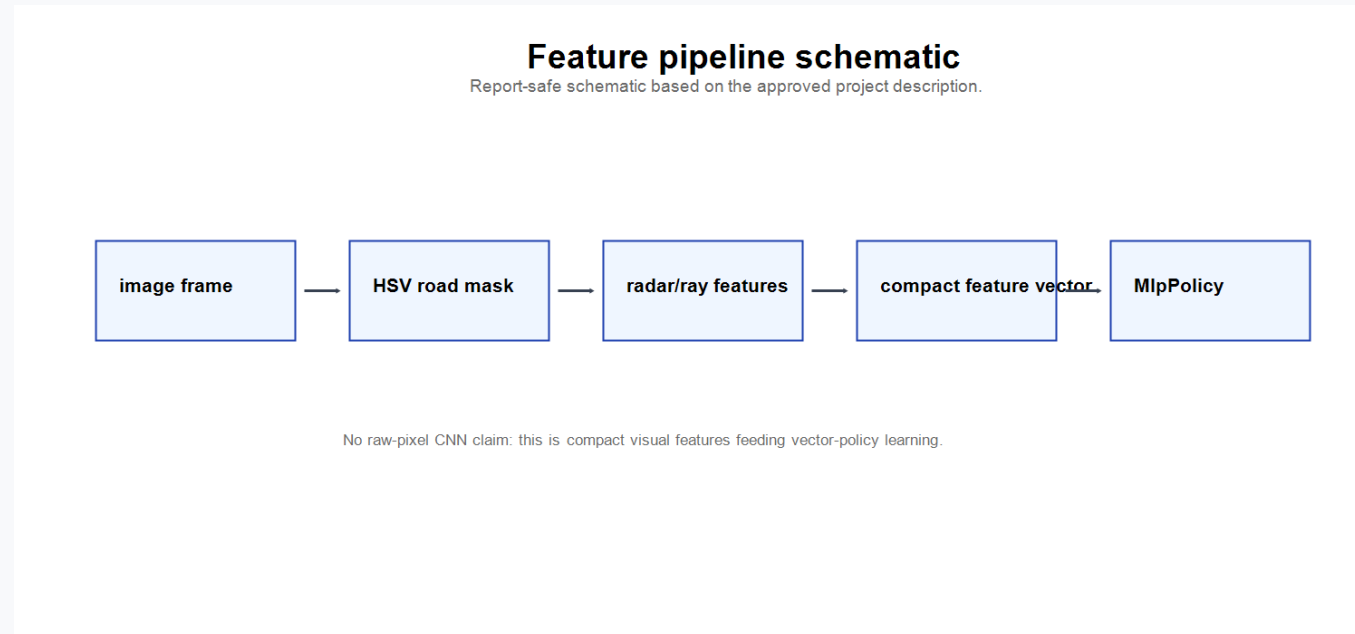
CarRacing-v3 environment

- Observation: rendered RGB frame
- Action: continuous steering, gas, brake
- Tracks: randomized across seeds
- Evaluation: saved logs, CSV tables, notebooks, and visual demos



The final claim is about a tested evidence package, not a solved environment.

Image processing pipeline



RGB frame -> crop/preprocess -> HSV road mask -> ray/radar features -> stacked observation -> PPO/SAC policy

The policy sees compact geometry, not raw pixels.

Compact feature representation

Road mask

HSV-based preprocessing separates road-like structure from background regions.

Ray/radar cues

Forward-biased distances summarize local drivable geometry.

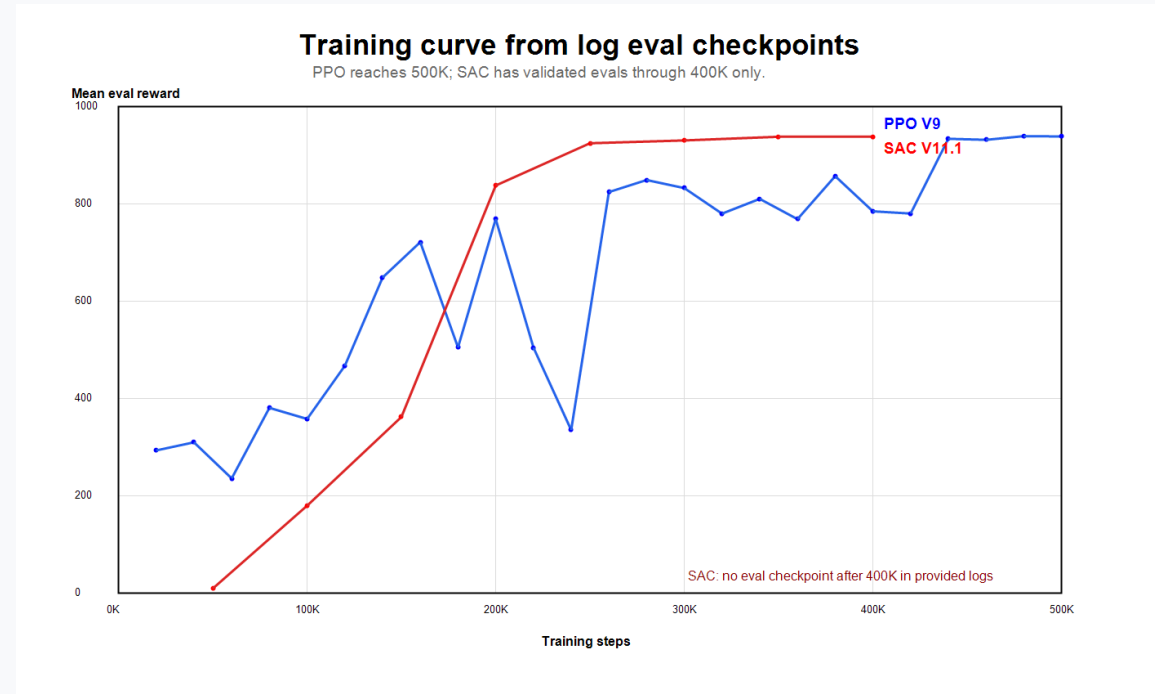
Stacked observation

Short-term context is carried through stacked compact features.

The representation is deliberately inspectable: failures can be related back to mask quality, local geometry, and recovery context.

PPO baseline (V9)

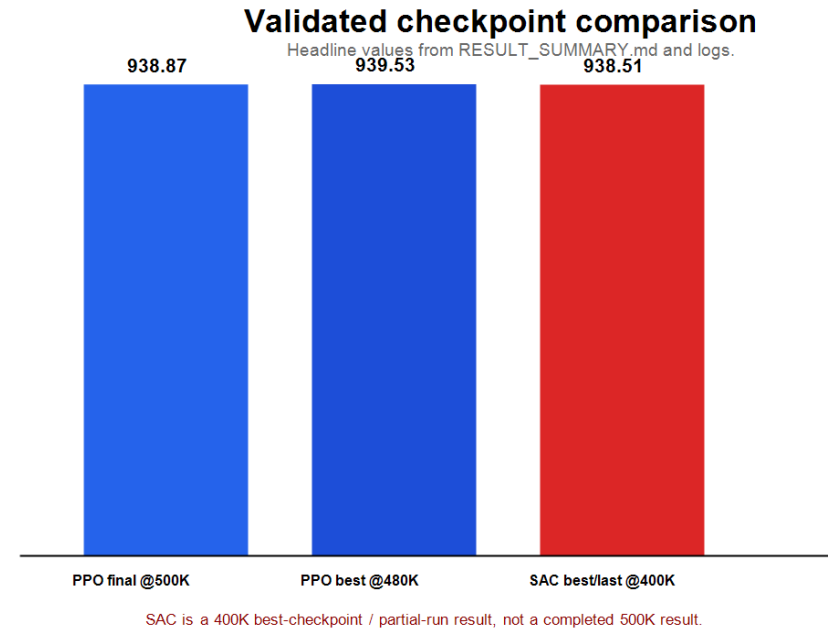
- Completed 500K baseline
- Final eval @500K: 938.87 +/- 7.86
- Best parsed eval: 939.53 +/- 4.09 @ 480,000
- Last observed step: 501,760



Evidence source: V2 result summary, PPO log, and CSV checkpoint table.

SAC fast-result branch (V11.1)

- Partial 400K best-checkpoint / fast-result branch
- Validated eval @400K: 938.51 +/- 4.88
- Last observed step: 418,697
- No validated eval checkpoint after 400K



SAC is not claimed to complete 500K, beat PPO, or provide a compute-equivalent comparison.

Demo videos and qualitative results



PPO higher-reward demo



PPO lower-reward demo



SAC higher-reward demo



SAC lower-reward demo

Videos are qualitative evidence extracted from notebook outputs. Quantitative claims come from logs and tables.

See: [media/videos/](#) and [media/video_manifest.csv](#)

Limitations

- SAC is a partial 400K checkpoint branch, not a completed 500K run.
- PPO and SAC are not compute-equivalent.
- No raw image-input CNN policy was trained in this V2 package.
- HSV-style preprocessing can be sensitive to visual changes.
- CarRacing-v3 is not claimed solved.

Reproducibility and GitHub structure

Final V2 folder

versions/2026-06-24-ppo-sac-feature-release/

Evidence

logs/, tables/, notebooks/, figures/, media/videos/

Final materials

report/Image_Processing_Project_V2_IEEE_Report.*
slides/Image_Processing_Project_Final_Presentation.*

The repository should be manually renamed to Image-Processing-Project before sharing the final link.

Conclusion and GitHub link

Image processing is the main contribution: compact road-mask and ray/radar features turn RGB frames into an interpretable policy input.

PPO baseline (V9): completed 500K, 938.87 +/- 7.86 @ 500,000

SAC fast-result branch (V11.1): partial 400K checkpoint, 938.51 +/- 4.88 @ 400,000

Final official repository after manual rename:

<https://github.com/UltraLamb/Image-Processing-Project>

Current repository before rename: <https://github.com/UltraLamb/CarRacingV3-PPO-RayFeatures>